

FUKUYAMA, Japan

WMO: 477670

Lat: 34.4469N

Lon: 133.2475E

Elev: 11

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 27.3	(c) 29.2	(d) 18.1	(e) 13.7	(f) 37.5	(g) 20.2	(h) 15.2	(i) 37.3	(j) 14.7	(k) 44.2	(l) 13.4	(m) 43.9	(n) 2.9	(o) 30	(p) 0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.6	(c) 93.3	(d) 78.0	(e) 91.5	(f) 77.8	(g) 89.7	(h) 77.3	(i) 79.8	(j) 89.4	(k) 78.9	(l) 88.3	(m) 78.0	(n) 87.1	(o) 7.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	141.9	84.4	76.3	137.2	83.8	75.4	133.0	83.3	43.4	89.4	42.4	88.3	41.5	87.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.4	(b) 11.5	(c) 9.8	(e) 23.6	(f) 96.2	(g) 2.5	(h) 2.0	(i) 21.8	(j) 97.6	(k) 20.4	(l) 98.7	(m) 19.0	(n) 99.8	(o) 17.1	(p) 101.2		
(a) 13.4	(b) 11.5	(c) 9.8	(e) 22.2	(f) 81.1	(g) 2.6	(h) 1.1	(i) 20.3	(j) 81.9	(k) 18.8	(l) 82.5	(m) 17.4	(n) 83.1	(o) 15.6	(p) 83.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	40.4	41.5	47.1	56.5	65.7	72.7	80.5	83.0	75.8	65.1	54.0	44.6
	DBStd	15.44	3.94	4.73	5.27	5.62	4.22	4.03	3.85	3.28	5.09	5.27	5.65	4.79
	HDD50	877	299	242	121	12	0	0	0	0	0	0	24	179
	HDD65	3301	763	658	553	259	42	1	0	0	0	62	330	633
	CDD50	4780	1	4	33	206	486	682	945	1024	774	469	145	11
	CDD65	1731	0	0	0	3	64	233	480	559	324	67	1	0
	CDH74	16011	0	0	0	23	362	1387	4875	6460	2610	294	0	0
	CDH80	5922	0	0	0	1	29	275	1922	2870	795	30	0	0
	WSAvg	3.7	3.9	3.8	4.1	4.0	3.8	3.6	3.8	4.0	3.7	3.4	3.3	3.7
	PrecAvg	47.7	1.7	2.1	3.6	3.9	5.5	6.9	7.8	3.9	5.2	3.4	2.5	1.6
(15) Precipitation	PrecMax	73.3	6.0	3.9	8.1	6.8	12.2	15.1	20.1	10.5	10.2	9.5	5.8	3.8
	PrecMin	28.6	0.3	0.7	1.7	1.2	1.2	0.9	1.4	0.6	1.8	1.0	0.5	0.1
	PrecStd	10.6	1.2	1.0	1.5	1.5	2.3	3.5	5.3	2.8	2.3	2.6	1.5	1.0
	PrecStd	10.6	1.2	1.0	1.5	1.5	2.3	3.5	5.3	2.8	2.3	2.6	1.5	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	61.1	68.0	77.1	83.1	89.2	94.6	96.2	92.6	83.5	71.9	62.0
		MCWB	48.5	53.0	55.2	61.2	65.8	75.2	78.2	78.2	77.2	73.0	62.4	53.3
	2%	DB	53.6	56.5	64.1	73.1	80.1	85.4	92.4	94.3	89.5	79.7	69.5	59.2
		MCWB	45.7	48.8	53.2	59.1	64.7	72.7	78.0	78.0	76.4	68.9	60.4	51.5
	5%	DB	51.3	53.8	60.7	70.2	77.9	83.1	90.6	92.6	87.1	77.2	66.9	56.7
		MCWB	44.1	46.4	51.5	58.3	63.7	71.2	77.7	77.9	75.2	66.5	58.8	49.5
	10%	DB	49.2	51.0	57.7	67.4	75.8	81.0	88.8	90.8	84.6	74.9	64.3	54.2
		MCWB	42.4	44.2	49.6	56.7	63.0	70.1	77.3	77.7	73.4	65.1	57.3	47.5
	0.4%	WB	51.2	55.0	58.3	66.1	70.8	77.0	81.3	81.0	79.0	74.7	65.5	57.0
		MCDB	54.1	59.5	63.0	70.4	76.6	85.6	90.0	91.5	88.1	80.5	68.6	59.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.5	50.4	55.6	63.0	68.4	75.3	79.8	80.0	77.7	71.7	63.0	53.0
		MCDB	51.4	54.7	60.5	68.3	74.2	82.5	88.8	90.1	85.8	78.0	67.1	57.4
	5%	WB	45.3	47.8	53.4	60.8	66.9	73.8	78.7	79.1	76.6	69.1	60.6	50.5
		MCDB	50.0	52.4	59.2	67.0	73.3	79.8	87.8	89.0	84.1	74.5	65.1	55.1
	10%	WB	43.2	45.2	50.9	58.9	65.4	72.5	77.8	78.4	75.5	66.7	58.2	48.3
		MCDB	48.4	50.1	57.1	65.3	72.6	78.1	86.6	87.9	82.5	72.7	63.3	53.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	16.7	17.9	18.4	17.8	14.0	13.3	14.6	14.7	16.5	17.1	16.4
		MCDBR	18.8	19.4	21.6	22.3	21.5	17.0	16.2	16.7	16.1	17.5	17.9	18.7
	5% WB	MCWBR	12.5	13.1	13.7	11.6	10.2	7.3	5.8	5.5	6.6	8.7	10.7	12.2
		MCWBR	17.3	16.6	18.2	16.3	15.4	12.5	14.9	15.1	13.8	13.9	14.9	16.1
	5% WB	MCWBR	12.6	12.4	13.2	10.0	9.0	6.1	5.8	5.7	5.9	8.2	10.4	12.0
		MCWBR	12.6	12.4	13.2	10.0	9.0	6.1	5.8	5.7	5.9	8.2	10.4	12.0
(40) Clear-Sky Solar Irradiance	taub		0.411	0.456	0.520	0.549	0.550	0.569	0.569	0.539	0.495	0.450	0.425	0.404
	taud		2.108	2.010	1.865	1.850	1.912	1.940	1.994	2.071	2.157	2.192	2.159	2.145
	Ebn at Noon		242	244	239	239	241	236	235	240	244	245	236	236
	Edn at Noon		40	49	61	65	62	60	56	51	45	40	37	36
(44) All-Sky Solar Radiation	RadAvg		738	936	1254	1535	1694	1505	1597	1640	1295	1078	801	648
	RadStd		50	75	96	136	158	142	209	162	132	104	66	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.58	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page